

E COMMERCE PRICE COMPARISON BOT

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In
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Certificate

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This is to certify that the work present in this Project entitled “E
COMMERCE PRICE COMPARISON BOT” has been carried out by
Deekshitha, Sucharitha, Sangeetha, Saisree under my/our supervision.
The work is genuine, original, and suitable for submission to the SRM
University – AP for the award of Bachelor of Technology/Master of
Technology in **School of Engineering and Sciences**.

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Abstract

The solution addresses the frequent problem that occurs when people shop online by looking for the lowest possible prices across multiple sites. The main challenge that enables shoppers to discover the lowest price for their desired item across multiple shopping websites. Users currently require separate browsing of Flipkart and Amazon along with Croma because those three websites operate as distinct destinations. The tool requires users to input the product name after which it launches searches on each shopping site and presents the pricing information. The required time together with significant effort make this process lengthy. We built a system which simplifies the process because we understood that customers needed help. A Python and Selenium WebDriver-based price comparison bot serves as the solution to this problem. The user interface prompts customers to provide their product selection and immediately activates a search among Flipkart, Amazon and Croma sites. The application automatically navigates between the Flipkart, Amazon and Croma websites consecutively. The program conducts keyword searches on each site eventually making the results data available to users. The bot functions as a genuine person through enhanced operating features to prevent websites from detecting its automated nature for person using the browser. The system operates with specific Chrome configuration elements and customized user-agent strings to look natural. The program deals with popups while it waits for page loading and uses tailored user-agent strings. The application applies effective methods to extract the accurate price information after the page finishes loading website. This program successfully adapts to minor structural alterations that appear on the websites. The bot identifies the product name or price presentation regardless of its location in the web page. The system maintains its attempts to locate the information through alternative techniques. The bot operates with reliability because of its design flexible. The system finishes its operations in seconds to provide the prices to users a lot of time. The bot achieves successful price comparisons for the majority of popular products. However, the website layout changes substantially or when the product becomes unavailable will prevent the system from

functioning properly. The bot system usually conducts price comparison during its operations correctly and quickly.

The bot offers online shopping customers time efficiency together with cost benefits through its automatic product price search feature. This system detects the best price available for each shopping item. It removes the need to Users benefit from opening alternative websites because it helps them search automatically. The project team plans to expand the platform by adding Reliance Digital and Tata CLiQ websites while also upgrading price accuracy through AI and developing mobile and web application interfaces. We plan to develop a mobile or web app through which more individuals can utilize the system also alongside upgrading price prediction accuracy via AI technology. The improved system would speed up online purchasing while delivering better speed and intelligence to the shopping process more convenient.

Introduction

Online buying through electronic and household item channels has become routine shopping behaviour for our present times. But one problem online shoppers often face is that people who shop online need to discover the precious deals available on their target products. The websites Flipkart and Amazon and Croma present identical products through their respective pricing structures. The various price points on websites lead to complications during comparison shopping for buyers seeking the ideal deal. Users must visit various websites to search for identical products. People must re-search the product multiple times while tracking various prices through a time-consuming process and effort. This project solves the problem through automated E Commerce Price Comparison using Python and Selenium WebDriver. The price comparison bot exists because of Python and Selenium WebDriver. The bot enables quick price comparison between e-commerce websites allowing users to find their optimum purchase deal. popular e-commerce websites like Flipkart, Amazon, and Croma. Through automation the bot both speeds up operations and helps people find ideal purchasing options effectively. The system accelerates the deal-

finding process thus allowing users to pick the best products. This program acts as a genuine user who navigates websites through standard browser functions allowing it to the websites that cannot detect the bot activities that make it unsafe for blocking. A user only requires inputting the details through the system. The user needs to enter the product name and the bot will complete a Flipkart search for the selected item.

The bot performs simultaneous price information collection operations on Flipkart along with Amazon and Croma.

site. This software handles routine problems related to login alerts and website loading while performing its work.

The price comparison process becomes smooth because the platform deals with login pop-ups, loading pages and different website layouts. process smooth and accurate.

Our project serves to deliver a user-friendly service for online shoppers who need better shopping efficiency.

This system assists online shoppers by minimizing their costs and time expenditure through improvements to their shopping process.

their shopping experience.

Methodology

Approach and Tools

This project involves building an automated price comparison bot that searches Flipkart, Amazon, and Croma for product prices. The bot mimics real-user behavior to avoid detection, automates the search process, and efficiently retrieves and displays price data for comparison.

Python – Used to build the bot.

Selenium – Helps control the browser.

Chrome Driver – Lets Selenium work with Chrome. **WebDriver Manager** Automatically sets up Chrome Driver **Time** – Adds wait time to avoid getting blocked.

Experimental Setup

1. **User Input:** User enters product name (e.g., "iPhone 14", "Laptop").
2. **Website Access:** Bot opens Flipkart, Amazon, and Croma in Chrome.
3. **Product Search:** It searches for the product on each site.
4. **Wait for Results:** Uses WebDriver Wait to wait for search results.
5. **Price Extraction:** Gets the first product's price using CSS/XPath.
6. **Result Display:** Shows all prices clearly for easy comparison

Data Processing Methods

Web Scarping: Bot collects price data from products listing.

Fallback Techniques: Tries backup methods if the first one fails.

Error Handling: Handles issues like popups, missing products, or layout changes.

Formatting Output: Displays the prices neatly in the console.

Software Requirements

1. **Python**
2. **Selenium WebDriver**
3. **Chrome Browser**
4. **Chrome Driver**
5. **WebDriver Manager**
6. **Time Library**

Implementation

1.Environment Setup: Installed Python, Selenium, and WebDriver Manager.

Set up Google Chrome and Chrome Driver for browser automation.

2.User Input: The bot takes a product name from the user (e.g., "iPhone 14").

3.Open E-Commerce Sites: The bot opens **Flipkart, Amazon, and Croma** using Chrome.

4.Search the Product: It enters the product name in each site's search bar and performs a search.

5.Wait for Page Load: Used WebDriver Wait to ensure the product listings load before continuing.

6.Extract Prices: The bot scrapes the price of the first matching product using CSS selectors or XPath.

If the first method fails, it tries an alternative one.

7.Display Result: Shows the prices from all three sites in the console for easy comparison.

Challenges Faced

Website layout changes frequently

Popups like login on Flipkart

Pages take time to load

Bot detection by websites

- Products not found or wrong match

Result and Analysis

After running the Price Comparison Bot with various product names (like "I phone 17", "Samsung phone ", "HP laptop "), the bot successfully:

- Took product input from the user.
- Searched the product on Flipkart, Amazon, and Croma.

- Extracted and displayed the first available price from each platform.
- Presented the prices in a clean, side-by-side comparison format in the console. **Insights**
- Value of Automation for price comparison
- Challenges in web scraping
- Error Handling and Resilience
- Importance of data formatting and consistency
- Selectors and dynamic waiting

Sample output

Discussion

The goal of this project was to create an easy tool for comparing products prices across different online stores, helping users find the best deals. The project achieves this by offering features like getting real-time prices ,comparing prices from Flipkart ,Amazon and Croma ,and allowing users to easily input the product name. The tool works offline, has a simple setup, and gives useful data through easy-to-understand text-based price comparisons. This project automates the price

comparison process, saving users time and helping them make better buying choices.

Limitations

1. Depends on the website's structure; any layout changes can break the scraper.
2. Limited to only three platforms: Flipkart, Amazon, and Croma.
3. Picks the first search result, which may not always match the exact product.

Doesn't compare product features, specifications, or customer reviews.

Conclusion

This project shows a simple and useful way to compare product prices from different online shopping websites. Using Selenium, it automatically searches and gets real-time prices from Flipkart, Amazon, and Croma. The tool is easy to use, runs from the command line, and needs very little input from the user. Even though it has some limitations—like depending on how websites are designed and not having extra features—it still saves users time by checking prices quickly. In the future, with some improvements, this tool can become even better and easier to use for smart online shopping.

Future Work

1.Add More E-Commerce Platforms

Include other popular websites like Reliance Digital, Tata CLiQ, and Paytm Mall for a wider price comparison.

2.Develop a GUI Interface

Create a simple and user-friendly graphical interface to make it easier for non-technical users.

3. Product Specification & Review Comparison

Fetch and compare product features, specifications, ratings, and customer reviews to help users make better choices.

4.Improve Product Matching Accuracy

Use smarter search logic or product IDs to ensure the correct product is compared across all platforms.

5.Add Sorting and Filtering Options

Allow users to filter results by price, brand, or rating and sort by lowest or highest price.

References

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(Used as one of the e-commerce platforms for price scraping)

4.Amazon India <https://www.amazon.in>

(Used as another source for product prices)

Croma Official Website

<https://www.croma.com>

(Used for fetching price details for comparison)

